

REQUIREMENT ANALYSIS

TABLES ATTRIBUTES:

1)BOOK DATA

ISBN, TITLE, AUTHOR_ID, PUBLISHER, LANGUAGE I.D, PUBLISHING DATE, EDITION_NUMBER, SIZE, PRICE, KEYWORD, SUBJECT, NO_PURCHASED_BOOKS, NO_SOLD_BOOKS

Here,

ISBN: It is the International Standard Book Number which is assigned to each separate edition and variation of a publication. Therefore, it is the Primary key of the table.

TITLE: it refers to the title of the book. (book name)

AUTHOR_ID: It refers to the number that is used to identify the author. It is the foreign key to the AUTHOR DATA TABLE

PUBLISHER: the person or company that issued the book.

LANGUAGE I.D: It contains a number between 0-4 which is used to identify which language the book was published in. It is a foreign key to the LANGUAGE_ID table.

PUBLISHING_DATE: Date of publication.

EDITION_NUMBER: It contains a numerical value which refers to which edition or variation of publication.

SIZE: Number of pages in the book

PRICE: Cost of the book in U.S dollars currency

KEYWORD: Special words associated with the book.

SUBJECT: It refers to the book genre:

- (a) Action and Adventure
- (b) Classics
- (c) Comic **Book** or Graphic Novel
- (d) Detective and Mystery
- (e) Fantasy
- (f) Historical Fiction
- (g) Horror
- (h) Science Fiction
- (i) Romance

NO_PURCHASED_BOOKS: It refers to the total number of books purchased by the bookstore or ever delivered by the supplier

NO_SOLD_BOOKS: It refers to the total number of books ever sold to the customers by the bookstore.

We can find the stock available for any book by simply subtracting:

$(NO_PURCHASED_BOOKS) - (NO_SOLD_BOOKS) = \text{Stock for a certain book}$

2)AUTHOR DATA

AUTHOR I.D, AUTHOR_NAME, LANGUAGE_ID

AUTHOR I.D: It is the numerical value that links to the author's name. It is the Primary key for the table.

AUTHOR NAME: It refers to the name of the author

LANGUAGE_ID: It contains a number between 0-4 which is used to identify which language the author's books are published in. It is a foreign key to the LANGUAGE_ID table.

3)LANGUAGE DATA

LANGUAGE_ID: It contains a number between 0-4 which is used to identify which language:

- (a) 0 = English
- (b) 1 = Spanish
- (c) 2 = French
- (d) 3 = Chinese Mandarin
- (e) 4 = others

LANGUAGE: It contains one of the above languages depending on the LANGUAGE_ID

4)SUBJECT DATA

SUBJECT_ID: It contains a number between 0-9 which is used to identify which book genre:

SUBJECT: The book genres available are:

- (a) Action and Adventure
- (b) Classics
- (c) Comic **Book** or Graphic Novel
- (d) Detective and Mystery
- (e) Fantasy
- (f) Historical Fiction
- (g) Horror
- (h) Science Fiction
- (i) Romance

4)CUSTOMER DATA

FIRST_NAME, LAST_NAME, CUSTOMER_ID, PASSWORD, ADDRESS, MOBILE_NUMBER, EMAIL_ID, BOOKS_ORDERED

Here,

FIRST_NAME: Customer's First name

LAST_NAME: Customer's Last name

CUSTOMER_ID: It is the login ID that customers can make to register and use to find information on their orders. It is the Primary Key for the Table and it is also a foreign key to the TRANSACTION HEADER TABLE.

PASSWORD: it is the password customers can use to login into their account.

ADDRESS: The customer's home address to deliver the books.

MOBILE_NUMBER: Mobile number of the customers to send information on their orders and provide notifications on newly arrived books and other suggested books.

EMAIL_ID: email address of the customers to send information on their orders and provide notifications on newly arrived books and other suggested books.

BOOKS_ORDERED: the number of books the customer has currently ordered

5)TRANSACTION HEADER TABLE

CUSTOMER_ID, TRANSACTION_ID, TRANSACTION_DATE, TRANSACTION_STATUS, FINAL_AMOUNT

CUSTOMER_ID: the login ID that customers use to place an order.

TRANSACTION_ID: Unique number to identify the specific transaction. It is the Primary key for the table. It is also the foreign key to the SHOPPING CART table.

TRANSACTION_DATE: Date of transaction

TRANSACTION_STATUS: It refers to whether the transaction is new, processing or closed.

FINAL_AMOUNT:

6)SHOPPING CART

TRANSACTION_ID, ISBN, ORDERED_QUANTITY, SHIPPED_QUANTITY, ITEM_STATUS, TOTAL_PRICE

TRANSACTION_ID: Unique number to identify the specific transaction.

ISBN: It is the International Standard Book Number which is assigned to each separate edition and variation of a publication.

ORDERED_QUANTITY: The number of copies of the same book the user wishes to purchase.

SHIPPED_QUANTITY:

BOOK_STATUS: It refers to whether each book on the transaction is new, processing or closed.

TOTAL_PRICE: It is the quantity multiplied with the book price.

In this table, TRANSACTION_ID and the ISBN act as a primary key.

7)COMMENTS

ISBN, CUSTOMER_ID, BOOK_SCORE, TEXT, FEEDBACK_RATINGS, USER_LEGITAMACY, FEEDBACK_DATE.

ISBN: It is the International Standard Book Number for which a user wishes to make a comment.

CUSTOMER_ID: It is the customer ID of a user wishes to make a comment.

BOOK_SCORE: It is a numerical score provided by the users for a particular book.

It can range from 0 to 10 where:

0 = terrible

2 = Not very good

4 = Average

6 = Not bad

8 = Good

10 = Masterpiece

TEXT: It is an optional short text user's can type as their comments.

USEFULNESS_RATINGS: Users can view a comment and provide a usefulness score on three levels:

(a) Useless

(b) Useful

(c) Very Useful

USER_LEGITAMACY: A user can declare any other user as trusted or not trusted.

FEEDBACK_DATE: The date a comment was made by a particular user.

CUSTOMER_ID and ISBN are used together to identify every comment by any user, and it is the primary key for the table.

8)MANAGER

FIRST_NAME, LAST_NAME, MANAGER_ID, PASSWORD, ADDRESS, MOBILE_NUMBER, EMAIL_ID

FIRST_NAME: Manager's First name

LAST_NAME: Manager's Last name

MANAGER_ID: It is the login ID that managers can make for other managers to register and use to find information on their customer's orders. It is the Primary Key for the Table.

PASSWORD: it is the password managers can use to login into their account.

ADDRESS: The manager's home address to deliver the books.

MOBILE_NUMBER: Mobile number of the managers to send notifications on newly arrived books.

EMAIL_ID: email address of the managers to provide notifications on newly arrived books.

9)CUSTOMER_PREFERED_LANGUAGES

CUSTOMER_ID, LANGUAGE_ID

10)CUSTOMER_PREFERED_SUBJECTS

CUSTOMER_ID, SUBJECT_ID

11)DELIVERY_DETAILS_TABLE

RECORD_NUMBER, RECEIVED_DATE, MANAGER_ID, ISBN, STOCK, SUPPLIER

RECORD_NUMBER: Unique number to keep track of all deliveries to the warehouse. It is the Primary key of the table.

RECEIVED_DATE: It is the date a particular delivery has been made and recorded.

MANAGER_ID: It is the manager ID of the manager present when the delivery has occurred and been recorded by the manager.

ISBN: It is the International Standard Book Number of the book that was delivered.

STOCK: The number of books that were delivered for each ISBN.

SUPPLIER: It is the name of the person or company that supplied the books to the warehouse.

FUNCTIONALITIES:

1)Customer Registration:

A new user has to provide the appropriate information as specified in the customer data table. He/she can pick a login name and a password.

Login names of the users must be unique to each other and are case insensitive. They should not contain any special characters and should not exceed a MINIMUM length of 5 characters and a MAXIMUM length of 8 characters.

Password of the user must have a minimum length of 10 characters and must contain at least 1 special character, 1 uppercase letter and 2 or more numbers.

Mobile number and email address must be provided in order to send notifications to users about their orders.

A customer can provide their preferred languages and their preferred subjects which can be later used for suggesting books.

book.

Every comment's date will be recorded in the COMMENT Table.

A numerical score between 0-10 where 0= terrible and 10=masterpiece can be provided by users for each book and an optional short text can also be made.

This numerical score is stored in BOOK_SCORE attribute in the COMMENT table.

A comment can only be changed by the customer who made it and only one comment per user per book is allowed.

2) Multiple Managers:

There can be multiple users who can access the full functionality of the system.

Only managers can create a new book while the user cannot.

Only a manager can register a new manager to the system.

There should be a superuser being the first manager when the system starts for the first time. Therefore, only a manager can register a new manager into the MANAGER table.

Only a manager can access the DELIVERY_DETAILS_HEADER TABLE to record the different deliveries made to the warehouse and users cannot access this table.

3)Ordering:

After registration, a new user can order one or more books by placing an order in the shopping cart.

The total amount of its order is reported to him/her.

A user may order multiple copies of a book one or more times.

A user can also see the status of their order in the shopping cart and are also provided with a transaction ID to keep track of their order.

This function is implemented using the SHOPPING CART table and users can see the quantity of each book they have ordered , how many of their books have been shipped and the status of the book.

4)New book:

The store managers may record the details of a new book and the number of books that have arrived in the warehouse in the DELIVERY_DETAILS_HEADER TABLE.

The store manager log in with their MANAGER_ID and record the date of delivery, ISBN of the books that were delivered and quantity of each book delivered.

They also record the supplier's name and a unique record number is generated for each delivery of books.

Only a manager can access the DELIVERY_DETAILS_HEADER TABLE to record the different deliveries made to the warehouse and users cannot access this table.

5. Arrival of more copies:

The store managers may change the stock level of a given book.

After a delivery of books has been made by a supplier to the warehouse, The store managers update the NO_PURCHASED_BOOKS in the BOOK DATA table by adding the quantity of the particular book received, i.e., by adding STOCK from DELIVERY_DETAILS_HEADER TABLE to the NO_PURCHASED_BOOKS in the BOOK DATA table.

In this way, a store manager can keep track of the stock level of any book by subtracting (NO_PURCHASED_BOOKS) and (NO_SOLD_BOOKS).

We can find the stock available for any book by simply subtracting:

$(\text{NO_PURCHASED_BOOKS}) - (\text{NO_SOLD_BOOKS}) = \text{Stock for a particular book.}$

6)Comments:

Users can record their comments for a

This functionality is implemented with the help of the COMMENT table and since every customer is allowed to make only one comment on every book, the CUSTOMER_ID and ISBN act as the primary key for the COMMENT table and can be used to uniquely identify every comment.

7. Usefulness ratings:

In this function, a user can view/access any customer's comment on any book/ISBN and provide a usefulness rating on that particular comment.

Users can access a comment and give it a usefulness score in three levels:

- (a) useless
- (b) useful
- (c) very useful.

A user should not be allowed to provide a usefulness rating for his/her own comments.

This is stored in the USEFULNESS_RATINGS attribute of the COMMENT table for every customer's comment on a particular book..

8) Trust recordings:

A user may declare zero or more other users as trusted or not trusted.

A user cannot declare a trust/not trusted rating for his/her own comments.

This is stored in the USER_LEGITAMACY attribute of the COMMENT table for every customer.

9)Book browsing:

User may search for books by asking conjunctive queries on authors, and/or publisher, and/or title words, and/or language and/or Subject.

The system allows the user to specify that the results are to be sorted:

- (a)by publish date
- (b)by edition
- (c) by the average numerical score of the comments
- (d) by the average numerical score of the trusted user feedbacks.
- (e) by their preferred languages and subjects.

We can sort results on publish date by ordering the PUBLISHING_DATE data in the BOOK data table.

Similarly, we can sort results on publish date by ordering the EDITION_NUMBER data in the BOOK data table by either ascending or descending order.

We can also sort the average numerical score of the comments and the average numerical score of the trusted users using the BOOK_SCORE and USER_LEGITAMACY data attributes respectively stored in the COMMENT table.

10) Useful Comments:

For a given book, a customer could ask for the top “n” most useful comments. The value n is user-specified (say 5 or 10). The usefulness of a comment is its average usefulness score.

11) Buying suggestions:

Like most e-commerce websites, when a user orders a copy of book A, the system gives a list of other suggested books. Book B is suggested if there is another user that bought both A and B, else another book is suggested if it falls in the same subject and language category.

The suggested books is sorted on the most quantity sold for each book (i.e., most popular first) among the users who also bought A.

This is achieved by filtering the books based on the subject and language category and then ordering the NO_OF_SOLD_BOOKS data in BOOK data table in descending order.

12) Degrees of separation:

Given an author X, a user may want to know the books written by authors who are 1-degree and 2-degree separated with X. The degree of separation is defined as follows: two authors A and B are 1-degree away if they co-authored at least one book together; they are 2-degrees away if there exists an author C who is 1-degree away from each of A and B **AND** A and B are not 1-degree way at the same time.

13) Book Statistics:

Every semester, the store manager would want:

- (a) The list of m most popular authors (in terms of copies sold this quarter)
- (b) The list of m most popular publishers (in terms of copies sold this quarter)
- (c) The list of m (say m = 10) most popular books (in terms of copies sold this quarter)

This can be implemented using multiple tables. For finding the top “m” most popular authors, we only have to order the NO_SOLD_BOOKS attribute in the BOOK DATA table in descending order for the particular period of the year and then link the corresponding AUTHOR_ID attributes with the AUTHOR_NAME attributes in the AUTHOR DATA table to list the popular author names.

We can similarly list the “m” most popular publishers in terms of copies sold this quarter.

14) **User awards:** The store manager also wants to give awards to the “best” users; Thus, the manager needs to know: • The top m most trusted users (the trust score of a user is the count of users trusting him/her minus the count of users not-trusting him/her)

- • The top m most useful users (the usefulness score of a user is the average usefulness of all his/her feedbacks combined).

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15)MANAGER AWARDS:

16) Most valuable customer

17)

E-R DIAGRAM: